

Functions of the ovary:

1. Ovaries play a critical role in both menstruation and conception. They produce eggs for fertilization and they make the hormones estrogen and progesterone.
2. These hormones are helpful in reproductive development and pregnancy.

3. During ovulation, one of the ovaries releases an egg. If a sperm fertilizes this egg, it leads to pregnancy.
4. Ovaries continue to release an egg each menstrual cycle until you reach menopause. During menopause, ovaries stop releasing eggs.

Functions of oviduct:

1. The main function of the oviduct or fallopian tube is to collect an ovum from the ovaries and transfer it to the uterus.
2. The eggs can be fertilized by sperm to become zygotes in the oviduct.
3. The fertilized egg then moves to the uterus, where it implants in the uterine lining.

Changes in the uterus when no fertilization takes place:

1. If no fertilization takes place there, the uterus wall which is unnecessary will break and will be removed through vagina along with mucosa and blood. This process is called menstrual cycle that may last for 2-8 days.
2. Menstruation starts during puberty and end at 40s or 50s. End of menstruation is termed as menopause.

Changes in the uterus when it has to receive a zygote:

1. The uterus undergoes numerous alterations in order to receive the zygote and produce a favorable environment for the commencement of pregnancy.
2. Every month, the uterus prepares itself to receive a fertilized egg/zygote. The inner uterine lining (endometrium) thickens and receives blood to nurture the embryo.

(a) Yellow - Dominant, Green - Recessive

(b) Round - Dominant, Wrinkled - Recessive

The number of chromosomes in an organism will be the same irrespective of whether it is male or female. As the male gametes in tobacco plant has 24 chromosome, the female gamete of the same plant also posses 24 chromosomes.

Zygote is the fertilized egg that results from the union of male and female gametes. As a result, the zygote will have 48 chromosomes.

Thus, **the correct answer is option (A).**

In females, the 23rd pair will contain two X chromosomes (XX). One X chromosome will be inherited from mother and the other X chromosome from father. In males, the 23rd pair of chromosomes will contain one X chromosome and one Y chromosome (XY).

Pancreas:

1. An organ that is located in the abdomen is called the pancreas.
2. It is crucial to the process of turning the food consumed energy for the body's cells.
3. The pancreas serves both as an exocrine and endocrine gland.
4. The exocrine gland is a gland that produces and releases bodily fluids such as saliva, perspiration, tears, milk, and digestive juices through a duct or other orifice to a body surface.
5. A gland that produces hormones that are immediately released into the circulation and go to all of the body's tissues and organs is called the endocrine gland.



6. Because it functions as both an endocrine and an exocrine gland, it is known as a mixed gland.
7. The pancreas produces both enzymes and hormones and aids in digestion.

1 Inadequate intake of iodine leads to insufficient production of these hormones, which adversely affect the muscle, heart, liver, kidney and the developing brain. This results in the disease states collectively known as Iodine Deficiency Disorders (IDD).

Functions of the placenta include:

Provides your baby with oxygen and nutrients. Removes harmful waste and carbon dioxide from your baby.

Produces hormones that help your baby grow. 4 Feb 2022